

The chosen support slice consists of almost-isometric generators

MathlibAnnex.FiniteRecovery.targetRawSupportMaximizer_good

theorem

Obtains one almost-isometric map from each point of the specified raw maximum slice.

Statement

Let $n = m + 1 > 0$. Use reference coordinates $V = \mathbb{R}^n$ with the sup norm and standard basis. The supplied continuous seminorm p has constants $a_p, b_p > 0$ satisfying

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in V).$$

Thus p is a norm; it can differ from the reference sup norm. With standard Lebesgue measure λ put $K_p = \{p \leq 1\}$ and $v_p = \lambda(K_p) > 0$. For a real linear map $A : V \rightarrow \mathbb{R}^N$, its matrix is $A_{ij} = (Ae_j)_i$. The vector $\Pi_n(A)$ has coordinates $\Pi_n(A)_I = \det(A_{I,*})$, one for each increasing n -tuple I of distinct rows. Define

$$\mathcal{G}_p^N = \left\{ \sigma v_p \Pi_n(A) : \begin{array}{l} A : V \rightarrow \mathbb{R}^N \text{ is real linear, } \sigma \in \{1, -1\}, \\ \|Ax\|_\infty \leq p(x) \quad (x \in V) \end{array} \right\}.$$

A base family $F = (\ell_i)_{i=1}^n$ consists of real linear functionals on V . Put $M_F = (\ell_i(e_j))$. Let $\mathcal{U}_p = \{F : \forall i, x, |\ell_i(x)| \leq p(x)\}$ and

$$D_p = \max_{F \in \mathcal{U}_p} |\det M_F|, \quad \mathcal{N}_{p,\eta} = \{F \in \mathcal{U}_p : D_p - \eta \leq |\det M_F|\}.$$

Fix $0 \leq \eta < D_p$, $\varepsilon > 0$, and one $K > 0$ such that $p(M_F^{-1}c) \leq K\|c\|_\infty$ for all $F \in \mathcal{N}_{p,\eta}$ and all $c \in V$. Set

$$\rho = \min(\varepsilon, 1)/(8K), \quad \mathcal{A}_K = \{c : 1/K \leq \|c\|_\infty \leq 1\}.$$

Take a finite $C_{\text{net}} \subset \mathcal{A}_K$ covering that annulus at sup-distance at most ρ ; its center a gives $c_a = a$. Fix $w > 0$ with

$$D_p \sum_{a \in C_{\text{net}}} \sum_i |(c_a)_i| < w\eta.$$

For a family of real linear functionals $s = (s_a)_{a \in C_{\text{net}}}$, call $Q = (F, s)$ admissible when $F \in \mathcal{U}_p$ and $|s_a(x)| \leq p(x)$ for all a, x . For this configuration put

$$\Phi_{w,c}(Q) = w \det M_F + \sum_a \sum_i (c_a)_i \det M_F[i \leftarrow s_a].$$

Here the replaced row is $(s_a(e_j))_j$. Put $N = n + |C_{\text{net}}|$. In minor space let I_0 select the base rows and $I_{i,a}$ select all base rows except i , then satellite a , and let e_I be the corresponding minor-coordinate unit vector. Define

$$\Omega_{w,c} = w e_{I_0} + \sum_a \sum_i (c_a)_i (-1)^{n-i} e_{I_{i,a}}.$$

Choose an admissible absolute maximizer Q_* of $\Phi_{w,c}$. Set $\theta = 1$ if $\Phi_{w,c}(Q_*) \geq 0$ and $\theta = -1$ otherwise, and define the linear functional $\ell(z) = \theta \langle \Omega_{w,c}, z \rangle$ (oriented support). If

$$z \in \mathcal{G}_p^N, \quad \ell(z) = \max_{u \in \mathcal{G}_p^N} \ell(u),$$

then one real linear $A : V \rightarrow \mathbb{R}^N$ satisfies

$$\forall x, \|Ax\|_\infty \leq p(x), \quad \forall x (p(x) = 1), \quad 1 - \varepsilon < \|Ax\|_\infty,$$

$$z = \sigma v_p \Pi_n(A) \quad \text{for some } \sigma \in \{1, -1\}.$$

Assumptions

The inputs are the continuous norm p with the displayed comparison bounds; $0 \leq \eta < D_p$; $\varepsilon > 0$; the common inverse bound $K > 0$; the stated finite net at radius ρ ; $w > 0$; the strict budget; and a point z in the displayed maximum slice of $\mathcal{G}_p^N$. All rows in the admissible configurations are real linear functionals. There is no assumption $\varepsilon < 1$.

Conclusion

One A has all three properties. For each unit vector the detecting satellite index may vary, while the map is fixed first.

Proof route

Recover the same generating map, transfer support maximality to absolute polynomial maximality, and use the finite-net estimate.

▼ Proof steps

1. **Recover the same map and compare both support values.** Membership in $\mathcal{G}_p^N$ supplies one real linear A and $\sigma \in \{1, -1\}$ with

$$\|Ax\|_\infty \leq p(x) \quad (x \in V), \quad z = \sigma v_p \Pi_n(A).$$

Extract the base and satellite rows of A as $Q = (F, s)$. With the base coordinates followed by the enumerated satellite coordinates,

$$A_{(F,s)}x = ((\ell_i(x))_{i=1}^n, (s_b(x))_{b \in C_{\text{net}}}).$$

Their individual bounds make Q admissible, and the extraction identity gives $A_{(F,s)} = A$. The normalized pairing identity and its oriented form yield

$$\begin{aligned} \ell(z) &= \sigma v_p \theta \Phi_{w,c}(Q) \\ &\leq v_p |\Phi_{w,c}(Q)| \\ &\leq v_p |\Phi_{w,c}(Q_*)|. \end{aligned}$$

For the reverse inequality, use the raw generator $v_p \Pi_n(A_{Q_*})$ with no extra sign. It belongs to $\mathcal{G}_p^N$, and

$$\begin{aligned}
\ell(v_p \Pi_n(A_{Q_*})) &= v_p \theta \Phi_{w,c}(Q_*) \\
&= v_p |\Phi_{w,c}(Q_*)| \\
&\leq \ell(z).
\end{aligned}$$

The middle equality is the support value at its chosen configuration; the last is precisely maximality of z . Hence both inequalities in the first chain are equalities. Since $v_p > 0$,

$$|\Phi_{w,c}(Q)| = |\Phi_{w,c}(Q_*)|.$$

This is raw support-to-configuration recovery for the original generating map A .

2. Apply finite-net detection to that configuration. For every fixed x with $p(x) = 1$, the required output is

$$\exists a \in C_{\text{net}}, \quad 1 - \varepsilon < |s_a(x)|.$$

Apply the finite-net estimate for an absolute maximizer with the same $p, \eta, \varepsilon, K, C_{\text{net}}, w$ and the same configuration Q . The inverse bound, net radius and strict budget are the stated inputs. Admissibility and absolute maximality of Q were established in Step 1. Only the detecting index a may depend on x ; the map A remains fixed.

3. Read the detecting functional as a coordinate of the fixed map. Fix x with $p(x) = 1$, and choose the index a supplied by Step 2. The map in Step 1 has the coordinate vector

$$Ax = A_{(F,s)}x = ((\ell_i(x))_{i=1}^n, (s_b(x))_{b \in C_{\text{net}}}).$$

Consequently its sup norm is

$$\|Ax\|_\infty = \max(\{|\ell_i(x)| : 1 \leq i \leq n\} \cup \{|s_b(x)| : b \in C_{\text{net}}\}).$$

The union is nonempty because $n > 0$. Since $a \in C_{\text{net}}$, the particular value $|s_a(x)|$ is one of the values in this maximum. This is the satellite-coordinate identity, applied to $Q = (F, s)$ and this same a, x , together with the defining property of the sup norm. Substituting the strict estimate from Step 2 therefore gives

$$1 - \varepsilon < |s_a(x)| \leq \|Ax\|_\infty \leq p(x) = 1.$$

The last inequality is the original contractivity from Step 1. The argument applies to every x with $p(x) = 1$; only a depends on x . Thus the original A is contractive and has the asserted strict lower bound on the unit sphere, while its original representation $z = \sigma v_p \Pi_n(A)$ is unchanged. These are precisely the three conditions in the good-generator RHS.

The sign θ belongs to the fixed functional ℓ , whereas σ represents the input raw generator. They have different roles. The raw set is compact and nonempty; its convex hull is the Plücker body. The hypothesis here concerns the raw maximum slice itself, not an arbitrary point of that convex hull.

Main citations

- Exact declaration and source
- oriented support

- compact
- nonempty
- extraction identity
- normalized pairing identity
- oriented form
- the support value at its chosen configuration
- raw support-to-configuration recovery
- the finite-net estimate for an absolute maximizer
- the good-generator RHS
- satellite-coordinate identity

Lean source signature (exact)

```
theorem targetRawSupportMaximizer_good
  {m : ℕ} (M : NormModel (m + 1)) {η ∈ weight : ℝ}
  (hη0 : 0 ≤ η) (hηD : η < detMax M) (hε : 0 < ε)
  (H : NearMaxInverseBound M η)
  (Cnet : FiniteCoefficientNet (n := m + 1) H.boundsConstant
    (satelliteRadiusNNReal ε H.boundsConstant hε H.boundsConstant_pos))
  (hweight : 0 < weight)
  (hgap : satelliteBudget M (centerValue Cnet) < weight * η)
  {z : PluckerCoord (m + 1)
    (positiveSatelliteAmbientDim m Cnet.centers)}
  (hz : z ∈ MathlibAnnex.NonemptyCompacts.maxSlice (rawPluckerCompact
    (N := positiveSatelliteAmbientDim m Cnet.centers) M)
    (orientedSatelliteSupport weight (centerValue Cnet)
      (maxSatelliteConfiguration M Cnet.centers
        weight (centerValue Cnet))))) :
  PluckerGeneratorGood M ε z
```

In the source	Mathematical meaning
$M : \text{NormModel } (m + 1); M.p$	The norm p on $V = \mathbb{R}^n$, with $n = m + 1$ and positive sup-norm comparisons.
$\text{PluckerCoord } (m + 1)$ $(\text{positiveSatelliteAmbientDim } m$ $\text{Cnet.centers})$	The real vector space with one coordinate for each increasing n -tuple of rows from $\{1, \dots, N\}$, where $n = m + 1$ and $N = n + C_{\text{net}} $.
$h\eta_0; h\eta_D; h\epsilon; \text{detMax } M$	Inputs $0 \leq \eta < D_p$ and $\epsilon > 0$; D_p is the attained absolute base-determinant maximum.

In the source	Mathematical meaning
<code>H : NearMaxInverseBound M η; H.boundConstant</code>	The same record holds $K > 0$ and the uniform bound $p(M_F^{-1}c) \leq K\ c\ _\infty$ on $\mathcal{N}_{p,\eta}$.
<code>Cnet; satelliteRadiusNNReal ε H.boundConstant hε H.boundConstant_pos</code>	The coefficient net on $\mathcal{A}_K$ at $\rho = \min(\varepsilon, 1)/(8K) > 0$. The proof arguments establish this radius rather than choosing further numbers.
<code>Cnet.centers; centerValue Cnet; positiveSatelliteAmbientDim m Cnet.centers</code>	The type <code>Cnet.centers</code> consists of the vectors belonging to the finite center set C_{net} . The function <code>centerValue Cnet</code> forgets only the membership proof: it is the inclusion $a \mapsto c_a = a$, and $N = n + C_{\text{net}} $.
<code>hweight; hgap</code>	$w > 0$ and the entire input $D_p \sum_a \sum_i (c_a)_i < w\eta$, for this coefficient family.
<code>rawPluckerCompact; maxSlice</code>	The raw set $\mathcal{G}_p^N$ consists of the vectors $\pm v_p \Pi_n(A)$ for real linear $A : V \rightarrow \mathbb{R}^N$ satisfying $\ Ax\ _\infty \leq p(x)$ for every x . <code>rawPluckerCompact</code> carries its compactness and nonemptiness; <code>maxSlice</code> selects its points maximizing the given linear functional.
<code>orientedSatelliteSupport weight (centerValue Cnet) (maxSatelliteConfiguration ...)</code>	The full $\ell(z) = \theta\langle \Omega_{w,c}, z \rangle$, with sign chosen from the fixed absolute maximizing configuration.
In the source	Mathematical meaning
<code>hz : z ∈ ... maxSlice ...</code>	Both $z \in \mathcal{G}_p^N$ and $\ell(z) = \max_{u \in \mathcal{G}_p^N} \ell(u)$.
<code>PluckerGeneratorGood M ε z</code>	Existence of one A with $\forall x, \ Ax\ _\infty \leq p(x)$, $\forall x, p(x) = 1 \Rightarrow 1 - \varepsilon < \ Ax\ _\infty$, and $z = v_p \Pi_n(A)$ or $z = -v_p \Pi_n(A)$.

▼ Exact content identity

Declaration: `MathlibAnnex.FiniteRecovery.targetRawSupportMaximizer_good`

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